5.3 Proving Triangle Congruence by SAS

Name _____ Period ____ Date _____

Geometry 5.3 · Student sheet

I can…

- identify the included angle between two sides.
- use SAS to decide whether two triangles are congruent.
- write a proof that uses SAS.

Vocabulary · fill in the missing word

Included angle — the angle formed by two sides of a _____

SAS Congruence Theorem — if two sides and the included angle of one triangle are congruent to two sides and the included angle of another, the _____ are congruent

Part 1 · Discovery: two sides and the angle between

Draw a 50° angle. Mark 4 cm on one ray and 6 cm on the other. Connect the endpoints.

	Third side	Other two angles	Matched my neighbor?
50° between the sides			
50° not between the sides			

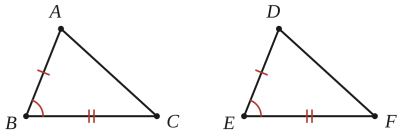
Part 2 · Notes and examples (fill in each note, then work the example)

Included angle: the angle _____ two sides (formed by them).

SAS Congruence Theorem: if two sides and the _____ angle of one triangle are congruent to two sides and the included angle of a second triangle, then the triangles are congruent.

Example 1

Can you prove the triangles congruent by SAS? If so, write the congruence statement.



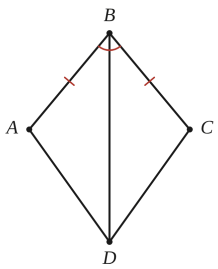
Example 2

$\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, and $\angle A \cong \angle D$. Can you use SAS? Explain.

Free parts: a shared side is congruent to itself by the _____ Property, and vertical angles are congruent.

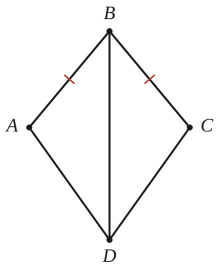
Example 3

$\overline{AB} \cong \overline{CB}$ and $\angle ABD \cong \angle CBD$. Are $\triangle ABD$ and $\triangle CBD$ congruent? Explain.



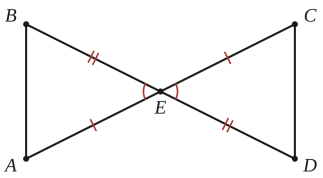
Example 4

Given: $\overline{AB} \cong \overline{CB}$ and $\overline{BD}$ bisects $\angle ABC$. Prove: $\triangle ABD \cong \triangle CBD$.



Example 5

Given: E is the midpoint of $\overline{AC}$ and of $\overline{BD}$. Prove: $\triangle AEB \cong \triangle CED$.



Part 3 · Partner practice

Partner 1

In $\triangle PQR$ and $\triangle XYZ$: $\overline{PQ} \cong \overline{XY}$, $\angle Q \cong \angle Y$, and $\overline{QR} \cong \overline{YZ}$. Are the triangles congruent? Which theorem?

Partner 2

Given: M is the midpoint of $\overline{AB}$ and of $\overline{CD}$ (the segments cross at M). Prove: $\triangle AMC \cong \triangle BMD$.